

CO2-AT2 Assignment

[← Back](#)

QUESTIONS

Q1Employee ID Normalization
System
10 marks**Q2**Bank Token Queue
Management System
10 marks**Q3**ATM PIN Security Validator
10 marks

3/3 answered

Q1 Employee ID Normalization System

10 marks

An HR department receives employee IDs in mixed formats such as emp1023. Create a Java program to:

Remove spaces using trim()

Convert the alphabetic prefix to uppercase using toUpperCase()

Join the prefix and number using concat()

Your Code

```
1 import java.util.*;
2 public class HRdepartment{
3     public static void main(String[] args){
4         Scanner sc = new Scanner (System.in);
5
6         String id = sc.nextLine();
7         id = id.trim();
8         String prefix = "";
9         String number = "";
10        for(int i = 0; i < id.length(); i++){
11            char ch =id.charAt(i);
12            if (Character.isLetter(ch)){
13                prefix = prefix + ch;
```

Input

anjali 1011

Output

ANJALI 1011



CO2-AT2 Assignment

← Back

QUESTIONS

Q1

Employee ID Normalization
System

10 marks



Q2

Bank Token Queue
Management System

10 marks



Q3

ATM PIN Security Validator

10 marks



3/3 answered

Q2 Bank Token Queue Management System

10 marks

A bank serves customers based on token order. Using Queue:

Add customer tokens using offer()

Serve the next customer using poll()

Display the upcoming customer using peek()

Your Code

```
1 import java.util.*;
2 public class Customers{
3     public static void main(String[] args){
4         Scanner sc = new Scanner (System.in);
5
6         Queue<Integer>q = new LinkedList<>();
7         int n = sc.nextInt();
8         for (int i = 0; i < n; i++){
9             q.offer(sc.nextInt());
10        }
11        System.out.println("Customers: " + q);
12        System.out.println("Served Customer: " + q.poll());
13        System.out.println("Upcoming Customer: " + q.peek());
14    }
15 }
```

Input

3
111
222
333

Output

Customers: [111, 222, 333]
Served Customer: 111
Upcoming Customer: 222



CO2-AT2 Assignment

← Back

QUESTIONS

Q1 ✓
Employee ID Normalization
System
10 marks

Q2 ✓
Bank Token Queue
Management System
10 marks

Q3 ✓
ATM PIN Security Validator
10 marks

3/3 answered

Q3 ATM PIN Security Validator

10 marks

A nationalized bank receives ATM PINs from customers through a kiosk. Develop a Java program that accepts the PIN as a string and performs the following operations:

Remove unwanted spaces using trim()

Validate that the PIN contains exactly 4 digits using matches()

Display the PIN length using length()

Your Code

```
1 import java.util.*;  
2 public class ATMPINs{  
3     public static void main(String[] args){  
4         Scanner sc = new Scanner(System.in);  
5         String pin = sc.nextLine();  
6         pin = pin.trim();  
7         if(pin.matches("\\d{4}")){  
8             System.out.println ("Valid PIN");  
9         }  
10        else{  
11            System.out.println ("Invalid PIN");  
12        }  
}
```

Input

97q2

Output

Invalid PIN
Length: 4